Deep-Learning Tool Detects Fetal Congenital Heart Disease (CHD) for Rural Coloradans

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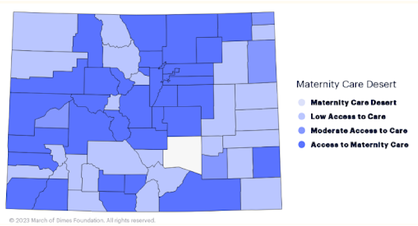
Motivation

- 40% of counties in Colorado do **not** have prenatal or delivery services
- long travel distances between towns, mountain roads, and hospital closures make care harder to reach
- about **1 in 100 babies** will need advanced heart care after birth
- need for a **free screening tool** that works closer to home

Maternity Care Deserts



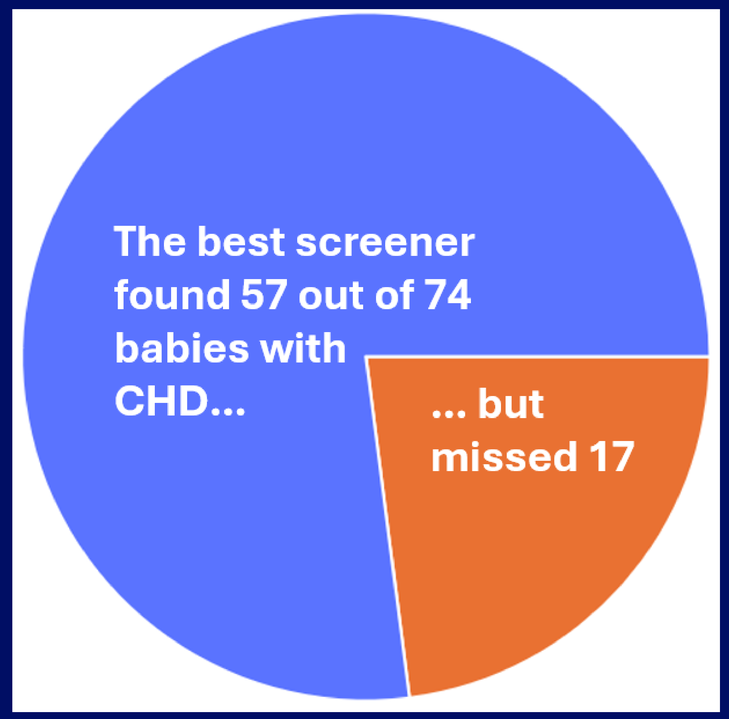
Existing Solutions

- many patients travel long distances for care, which takes time and money
- screening often needs **highly trained specialist**
- existing AI-screening tools are **expensive and not publicly available**
- current models focus on overall accuracy instead of **avoiding missed heart problems**

Idea

Automated ultrasound screening tool deployed by traveling sonographers to support earlier screening decisions closer to where patients live.

If you lived in a maternity care **desert,** would you **trust AI** to find problems with **your pregnancy?**



Take a picture to download the full proposal

Approach & Design Decisions

- traveling sonographers screen patients and refer higher-risk cases to fetal heart specialists
- trained our model using labeled ultrasound images and clinical data from the CARDIUM dataset and FetalCLIP
- tested three training methods (BCE, weighted BCE, focal loss functions) designed to reduce **missed heart problems** when the condition is rare
- A loss function measures how wrong its predictions are so it can learn to make better ones next time.
- prioritized higher sensitivity to support earlier referral, delivery planning, and access to specialized care

Key Findings

BCE detected 80% of CHD cases and required extra screening for about half of patients without CHD. Weighted BCE detected the most cases (89%) but required the most additional screening (58%). Focal loss detected slightly fewer cases (77%) while reducing extra screening to 31%.

1 in 3 patients without CHD would be flagged for extra screening